

Evaluating OpenCap for Collecting and Analyzing Human Movement with Smartphones

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BACKGROUND

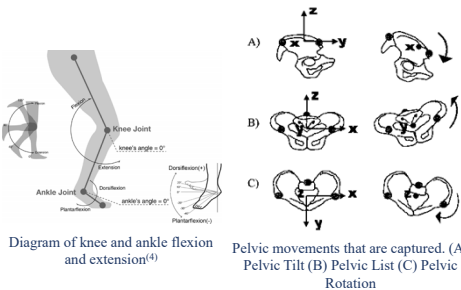
- Current methods for motion capture are costly, time consuming and advanced experience is required to collect and analyze the biomechanical motion data.
- OpenCap was developed as an open-source software for collecting 3D human movements using videos captured from smartphones¹.
- The web application allows users to sync videos from 2 to 4 iPhone cameras and uses computer vision-based pose detection to output the 3D movements of the human body.
- OpenCap may allow researchers, physical trainers, coaches and physical therapists to record their clients quickly and share the video analysis to get a visual coaching on what they are doing incorrectly.
- Educators can also use OpenCap in their classes to help students understand biomechanics of human movements.

OBJECTIVES

The goal of this project is to explore how OpenCap can be used for evaluating human movements in sports, education, aging or ergonomics. Comparisons were made with the current standard equipment used in biomechanics labs (Vicon Motion Capture). Participants simulated movements that would be typical done with a physical therapist or athletic trainer, including; Functional Movement Screens (FMS), throwing, walking, running, jumping and cutting. 3D motion of the body and the kinematics of the joints were calculated from both methods.



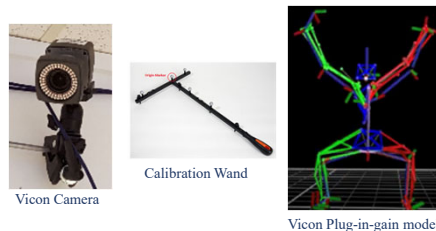
Typical functional movement screen activities that could be quantified with motion capture.



MATERIALS & METHODS

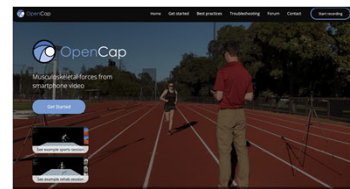
Vicon Motion Capture Procedure:

- The setup uses 10 cameras, 5-5 on each side to cover the capture volume zone from all the sides giving a 3D view².
- Nexus software was used to collect all the data from the cameras and the force plate.
- Cameras were calibrated with a five marker wand that has markers on it.
- Vicon Plug-in-gait biomechanical model is used for data analysis of markers on the subject to 3D joint kinematics.

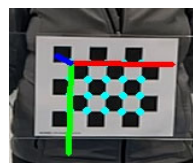


OpenCap Procedure:

- At least 2 iOS devices (2018 or later) were mounted on tripods with phone holders and their cameras pointed at the capture volume.
- A printed checkerboard from OpenCap's web page was used for the initial capture volume calibration³.
- Recorded videos are uploaded to OpenCap's server for processing using video pose detection algorithms. The processed motion data includes the subject's 3D position and joint kinematics.



OpenCap Web Application



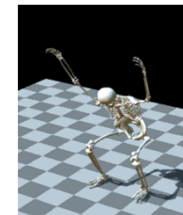
Calibration Board

RESULTS

- Kinetics parameters were capture from both systems included pelvic translations and rotations, hip flexion, hip adduction, hip rotation, knee flexion, ankle flexion, ankle inversion, trunk rotations, shoulder rotations, elbow flexion, wrist flexion.



Video Analysis of Deep Squat

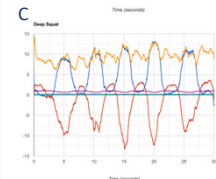
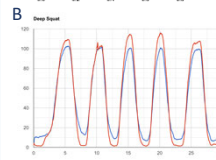


OpenCap 3D Model

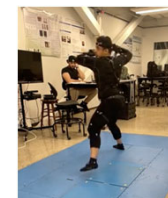


Kinematic graphs for left and right knee flexion are similar during the deep squat FMS activity.

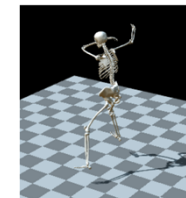
A: Vicon Motion Capture (one squatting cycle shown)
B: OpenCap (five squatting cycles)



C: Pelvis kinematics during the deep squat. As the subject reaches the squat the position the pelvis tilts forward which causes the pelvis to translate rearwards.



Video Analysis of Pitching



OpenCap 3D Model

During baseball pitching the OpenCap 3D model frame rate was not fast enough to accurately track the subject motions.

CONCLUSION

- OpenCap demonstrated that it can gather the same data as the motion capture system. There are certain things that the system is un able to do.

PROS	CONS
- Inexpensive - Short setup time - Portable - Video analysis	Video frame rate during fast paced movement

Who can benefit from it?

- Physical Therapist
- Physical Trainers
- Educators

FUTURE WORK

Things to Consider

- Frame rate
- Number of cameras
- Size of capture volume
- Activity being performed
- Data analysis on web application
- Work with other devices devices iOS

REFERENCES

- Uhlrich et al. (2022) OpenCap: 3D Human Movement Dynamics from smartphone videos. ArXiv preprint.
- Vicon Motion Capture Systems <https://www.vicon.com/>
- OpenCap <https://www.opencap.ai/>
- Kim et al. (2019) Kinematic on Ankle and Knee Joint of Post-Stroke Elderly Patients by Wearing Newly Elastic Band-Type Ankle-Foot Orthosis in Gait. Clinical Interventions in Aging.

ACKNOWLEDGEMENTS

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